

Algebra II with Trigonometry

Chapter 5.2

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:

[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Solutions

1. Evaluate exactly:

$$\frac{\sin 15^\circ + \cos 15^\circ}{\sin 15^\circ \cos 15^\circ}.$$

Use

$$\sin x + \cos x = \sqrt{2} \cos(x - 45^\circ),$$

so at $x = 15^\circ$,

$$\sin 15^\circ + \cos 15^\circ = \sqrt{2} \cos 30^\circ = \sqrt{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{2}.$$

Also,

$$\sin 15^\circ \cos 15^\circ = \frac{1}{2} \sin 30^\circ = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Therefore

$$\frac{\sin 15^\circ + \cos 15^\circ}{\sin 15^\circ \cos 15^\circ} = \frac{\frac{\sqrt{6}}{2}}{\frac{1}{4}} = 2\sqrt{6}.$$

$$\boxed{2\sqrt{6}}$$

2. A right triangle has acute angle θ with

$$\tan \theta + \cot \theta = 4.$$

Find the exact value of $\sin \theta + \cos \theta$.

We have

$$\tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta}.$$

Hence

$$\frac{1}{\sin \theta \cos \theta} = 4 \quad \Rightarrow \quad \sin \theta \cos \theta = \frac{1}{4}.$$

Now

$$(\sin \theta + \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 1 + 2 \cdot \frac{1}{4} = \frac{3}{2}.$$

Since θ is acute, $\sin \theta + \cos \theta > 0$. Thus

$$\sin \theta + \cos \theta = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}.$$

$$\boxed{\frac{\sqrt{6}}{2}}$$

3. Two observers stand on the same straight road on opposite sides of a tower. Their distances from the tower differ by 40 m, and the angles of elevation to the top are 45° and 30° . Find the height of the tower.

Let the tower height be h . Let the distances of the two observers from the tower be x and y , with angles of elevation 45° and 30° respectively.

Since they are on opposite sides, the distances are simply horizontal distances from the tower, and we are given

$$|y - x| = 40.$$

From the 45° observer,

$$\tan 45^\circ = \frac{h}{x} = 1 \quad \Rightarrow \quad x = h.$$

From the 30° observer,

$$\tan 30^\circ = \frac{h}{y} = \frac{1}{\sqrt{3}} \quad \Rightarrow \quad y = h\sqrt{3}.$$

Because $\sqrt{3} > 1$, we have

$$y - x = h(\sqrt{3} - 1) = 40.$$

So

$$h = \frac{40}{\sqrt{3} - 1} = \frac{40(\sqrt{3} + 1)}{3 - 1} = 20(\sqrt{3} + 1).$$

Thus the height of the tower is

$$\boxed{20(\sqrt{3} + 1) \text{ m}}.$$

4. Without a calculator, determine the exact value of

$$(\sec 30^\circ - \tan 30^\circ)(\sec 30^\circ + \tan 30^\circ) + (\csc 45^\circ - \cot 45^\circ)^2.$$

First term:

$$(\sec 30^\circ - \tan 30^\circ)(\sec 30^\circ + \tan 30^\circ) = \sec^2 30^\circ - \tan^2 30^\circ.$$

Using the identity $\sec^2 \theta - \tan^2 \theta = 1$, this equals

$$1.$$

Second term:

$$\csc 45^\circ = \sqrt{2}, \quad \cot 45^\circ = 1,$$

so

$$(\csc 45^\circ - \cot 45^\circ)^2 = (\sqrt{2} - 1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}.$$

Adding,

$$1 + (3 - 2\sqrt{2}) = 4 - 2\sqrt{2}.$$

$$\boxed{4 - 2\sqrt{2}}$$

5. In a right triangle, the altitude from the right angle to the hypotenuse divides the hypotenuse into segments of lengths 1 and 4. Find the exact value of the triangle's perimeter.

Let the right triangle have hypotenuse c , and let the altitude to the hypotenuse divide it into segments 1 and 4. Then

$$c = 1 + 4 = 5.$$

A standard right-triangle altitude theorem says: if the hypotenuse is divided into segments m and n , then the legs satisfy

$$a^2 = cm, \quad b^2 = cn.$$

Here $m = 1$, $n = 4$, $c = 5$. Thus

$$a^2 = 5 \cdot 1 = 5 \quad \Rightarrow \quad a = \sqrt{5},$$

$$b^2 = 5 \cdot 4 = 20 \quad \Rightarrow \quad b = 2\sqrt{5}.$$

Therefore the perimeter is

$$a + b + c = \sqrt{5} + 2\sqrt{5} + 5 = 3\sqrt{5} + 5.$$

$$\boxed{5 + 3\sqrt{5}}$$

6. A balloon is vertically above a straight road. The angles of depression to two mileposts on the same side of the

balloon are 45° and 30° . How high is the balloon above the road?

Let the balloon's height above the road be h . Let the nearer and farther mileposts be at horizontal distances x and y from the point on the road directly below the balloon.

Since the angles of depression equal the corresponding angles of elevation, we have

$$\tan 45^\circ = \frac{h}{x} \Rightarrow x = h,$$

and

$$\tan 30^\circ = \frac{h}{y} \Rightarrow y = h\sqrt{3}.$$

The two mileposts are consecutive, so they are 1 mile apart. Since they are on the same side,

$$y - x = 1.$$

Hence

$$h\sqrt{3} - h = 1 \quad \Rightarrow \quad h(\sqrt{3} - 1) = 1$$

and therefore

$$h = \frac{1}{\sqrt{3} - 1} = \frac{\sqrt{3} + 1}{2}.$$

So the balloon is

$$\boxed{\frac{\sqrt{3} + 1}{2} \text{ miles}}$$

above the road.